ISLAMIC UNIVERSITY OF TECHNOLOGY

Course Outline and Course Plan

Name of the Teacher	Shahriar Ivan	**Position**	Assistant Professor
Department	CSE	**Program**	B.Sc. Engineering in SWE
Course Code	CSE 4107	**Course Title**	Structured Programming I
Academic Year	2024-25	**Semester**	Winter
Contact Hours	3.00	**Credit Hours**	3.00
Textbooks and Reference books	1. C Programming - A Modern Approach 2. Teach Yourself C 3. Programming in ANSI C	**Author of the Books**	1. K. N . King 2. Herbert Schildt 3. E. Balagurusamy
Prerequisites	A good grasp on College Algebra and Problem Solving		
Corequisites			
Course Homepage	Google Classroom Code: czyfcbci		
Teaching Methods/ Approaches	Lecture Demonstration Problem Solving		
Teaching Aids	Multimedia and OHP Board and Marker Video Conferencing Tool and Graphics Tablet		

Course Assessment Method						
Attendance (0%)	Quiz/Viva (20%) (Best 3 out of 4)				Mid Exam (40%)	Final Exam (40%)
	1st Quiz	2nd Quiz	3rd Quiz	4th Quiz	Week/ Date	Week/ Date
	Week/ Date	Week/ Date	Week/ Date	Week/ Date		
	4th Week	7th Week	10th Week	14th Week	8-9th Week	17-18th Week

Grading Policy					
Marks out of 100	Letter Grade	Grade Point	Marks out of 100	Letter Grade	Grade Point
80 - 100	A+	4.00	55 - 59	B-	2.75
75 - 79	A	3.75	50 - 54	C+	2.50
70 - 74	A-	3.50	45 - 49	C	2.25
65 - 69	B+	3.25	40 - 44	D	2.00
60 - 64	B	3.00	0 - 39	F	0.00

Course Contents

Intro to C Programming Language: Introduction to programming languages, Standards, Basic I/O
Programming Style: Formal aspects of language syntax and semantics
Basic and Advanced Language Features: Operators, Conditional Statements, Loops, Functions, File I/O etc.
Problems Solving Techniques: Define the problem, List possible solutions, Evaluate methods, Create an implementation plan, Communicate the solution
Programming Environments and Tools: Use of different programming development environments and tools

Course Objectives

The course aims to provide the student with:
1. Knowing the basics of Computer Programming Language
2. Exploring standard I/O, Conditional statements, Iteration statements
3. Learning about Data types and Variables
4. Using Arrays and Functions in different scenarios
5. Understanding Pointers and Dynamic Memory Allocation
6. Acquiring knowledge about Structures and Unions
7. Getting insights on File I/O and Low-Level Programming

Mapping with CO, PO, and Bloom's Taxonomy

CO No.	Course Outcomes (CO) Statement	Levels of Bloom's Taxonomy	Matching with Program Outcome (PO)
CO1	Understand different programming language concepts	C1 (Remember)	PO1
CO2	Analyze which application scenario requires which Programming language features	C2 (Understand)	PO2
CO3	Apply the programming language expertise to solve complex engineering problems	C3 (Apply)	PO1

Weekly Plan for Course Content and Mapping with CO		
Weeks	**Topics**	**COs**
1	Introduction, Origin, Standards, Comments, Basic I/O, Identifiers, Keywords	CO1
2	Fundamentals of C	CO1
3	The Printf Function, The Scanf Function	CO1
4	Arithmetic Operators, Assignment Operator, Increment and Decrement Operators, Expression Evaluation	CO1, CO2
5	Logical Expressions, The If Statement, The Switch Statement	CO1, CO2
6	The While Statement, The Do Statement, The For Statement, Exiting from a Loop, The Null Statement	CO1, CO2
7	Integer Types, Float Types, Character Types, Type Conversion	CO1
8	Mid Exam	
9		
10	One-Dimensional Arrays, Multidimensional Arrays, Variable-Length Arrays, Function Definition, Calling Functions, Declarations, Arguments, The return Statement, Termination, Recursion	CO1, CO2, CO3
11	Pointer Variables, Assignment, Pointers as Argument, Pointers as Return values	CO1, CO2, CO3
12	Pointer Arithmetic, Using Pointers for Array Processing, Using Array name as Pointer, Pointers and Multidimensional Arrays	CO1, CO2, CO3
13	String Literals, String Variables, Reading and Writing Strings, Accessing Characters in a String, C String Library, Arrays of Strings	CO1, CO2, CO3
14	Structure Variables, Structure Types, Nested Arrays and Structures, Unions, Enumerations, Streams, File Operations, File I/O, File Positioning	CO1, CO2, CO3
15	Dynamic Storage Allocation, Dynamically Allocated Strings, Dynamically Allocated Arrays, Deallocating Storage, Linked List, Pointers to Pointers	CO1, CO2, CO3
16	How Preprocessor works, Preprocessor Directives, Macro Definitions, Conditional Compilation, Bitwise Operations, Bit-Fields in Structures	CO1, CO2, CO3
17	Final Exam	
18		
19		

Program Outcomes	
PO 1	**Engineering Knowledge:** Apply the knowledge of mathematics, science, engineering funda mentals and an engineering specialization to the solution of complex engineering problems.
PO 2	**Problem Analysis:** Identify, formulate, research and analyze complex engineering problems and reach substantiated conclusions using the principles of mathematics, the natural sciences and the engineering sciences.
PO 3	**Design/Development of Solutions:** Design solutions for complex engineering problems and design system components or processes that meet the specified needs with appropriate consideration for public health and safety and of cultural, societal and environmental concerns.
PO 4	**Investigation:** Conduct investigations of complex problems, considering experimental design, data analysis and interpretation and information synthesis to provide valid conclusions.
PO 5	**Modern Tool Usage:** Create, select and apply appropriate techniques, resources and modern engineering and IT tools, including prediction and modeling, to complex engineering activities with an understanding of their limitations.
PO 6	**The Engineer and Society:** Apply reasoning informed by contextual knowledge to assess societal, health, safety, legal and cultural issues and the consequent responsibilities relevant to professional engineering practice.
PO 7	**Environment and Sustainability:** Understand the impact of professional engineering solutions in societal and environmental contexts and demonstrate the knowledge of and need for sustainable development.
PO 8	**Ethics:** Apply ethical principles and commit to the professional ethics, responsibilities and the norms of the engineering practice.
PO 9	**Individual Work and Teamwork:** Function effectively as an individual and as a member or leader of diverse teams and in multidisciplinary settings.
PO 10	**Communication:** Communicate effectively about complex engineering activities with the engineering community and with society at large. Be able to comprehend and write effective reports, design documentation, make effective presentations and give and receive clear instructions.
PO 11	**Project Management and Finance:** Demonstrate knowledge and understanding of engineering and management principles and apply these to one's work as a team member or a leader to manage projects in multidisciplinary environments.
PO 12	**Life-Long Learning:** Recognize the need for and have the preparation and ability to engage in independent, life-long learning in the broadest context of technological change.

Mapping of COs and POs [Correlation level 1 for low, 2 for moderate and 3 for high]												
Course Outcomes	PO 1	PO 2	PO 3	PO 4	PO 5	PO 6	PO 7	PO 8	PO 9	PO 10	PO 11	PO 12
CO 1	3											
CO 2	2											

CO 3	2	3										

Student Consulting Hour:

- Thursday, 10 a.m. - 12 p.m.
- Saturday, 10 a.m. - 11 p.m. (on request)

Instruction Contact Details:

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